

Lezione 11 esercizi 2

venerdì 17 aprile 2026 17:18

1) numeri LD

$$t = \left\lfloor \frac{n(R - 0.34 \log n - 0.71)}{-0.73 \log n - 0.296} \right\rfloor$$

$$M: 2000, R = 3/4$$

$$t = 74$$

$$\text{Se } p_b = 2 \cdot 10^{-2} \Rightarrow \binom{2000}{i}$$

$$\mu = n \cdot p = 40$$

$$\sigma^2 = np(1-p) = 39.2 \Rightarrow \sigma = 6.3$$

$$1. p_p = \sum_{i=0}^{75} \binom{2000}{i} p_b^i (1-p_b)^{2000-i}, p_b = 2 \cdot 10^{-2}$$

troppo lungo !!!

sostituisco con Gaussiana! $\leftarrow np > 10$!

$$\Rightarrow \int_{35}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{2\sigma^2}} dx \quad \text{dopo cambio } \Rightarrow$$

$$\Rightarrow \int_{\frac{35}{\sqrt{np(1-p)}}}^{\infty} \frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}} dx = \frac{1}{2} e^{-\frac{\left(\frac{35}{6.3}\right)^2}{2}} = 9.9 \cdot 10^{-8}$$

$$p_b = \frac{d}{n} p_{ep} = 7.3 \cdot 10^{-9} \quad (d = 2 \text{ km})$$

2) $t \times r(t) + x'(t) = \frac{d\delta(t)}{dt}$

↓

Fourier

$$\frac{1}{(-j\pi f)} \frac{d\chi(f)}{df} + (j2\pi f) \chi(f) = j2\pi f$$

↓

$$\frac{d\chi(f)}{df} + (4\pi^2 f) \chi(f) = 4\pi^2 f$$

$$\frac{d\chi(f)}{df} + 4\pi^2 f \chi(f) = 0 \Rightarrow \frac{d\chi(f)}{df} = -4\pi f \chi(f) \Rightarrow$$

$$\Rightarrow \frac{d\chi(f)}{\chi(f)} = -4\pi f df \Rightarrow$$

$$\Rightarrow \log \chi(f) = -2\pi f^2 + C \Rightarrow \chi(f) = k e^{-2\pi f^2}$$

omogenea

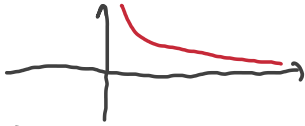
Nel caso lineare con:

... ..

$$y = e^{-\gamma \pi^2 f^2} \left[\int p(\lambda) e^{\gamma \pi^2 \lambda^2} d\lambda + k \right] \Rightarrow$$

$$\begin{aligned} X(f) &= e^{-\int \gamma \pi^2 f^2 d\lambda} \left[\gamma \pi^2 \int e^{\gamma \pi^2 f^2} d\lambda + k \right] \\ &= e^{-2\pi f^2} \left[2\pi f e^{2\pi f^2} + k \right] = 2\pi f + k e^{-2\pi f^2} \end{aligned}$$

$$3) \lambda(t) = t^{-1/2} \quad \forall t \Rightarrow \frac{1}{\sqrt{t}}$$



$$= \int_0^a t^{-1/2} dx = \frac{t^{-1/2+1}}{-1/2+1} = \frac{t^{1/2}}{1/2} \Big|_0^a = \frac{a^{1/2}}{1/2}$$

$$\frac{d\lambda(t)}{dt} = -\frac{1}{2} t^{-3/2} = -\frac{1}{2} \frac{t^{-1/2}}{t} =$$

$$= -\frac{1}{2} \lambda(t)$$

$$\frac{d}{dt} \left[\frac{1}{\sqrt{t}} \right] = -\frac{1}{2} \frac{1}{\sqrt{t}}$$

$$\Rightarrow \lambda(t) + t \frac{d\lambda(t)}{dt} = \frac{1}{2} \lambda(t) \Rightarrow$$

$$-\frac{1}{2} \frac{d\lambda(t)}{dt} = -\frac{1}{2} \lambda(t) \Rightarrow$$

$$\frac{d\lambda(t)}{\lambda(t)} = -\frac{1}{2} \frac{dt}{t}$$

$$\Rightarrow \log \lambda(t) = -\frac{1}{2} \log t + c$$

$$\Rightarrow \lambda(t) = \frac{k}{\sqrt{t}}$$

$$4) \text{ LTI } \lambda(t) = \frac{1}{t^2-1}, \quad t > 0, \quad h(t) = t, \quad 0 < t < 3$$

$$y(t) = \int \lambda(\tau) h(t-\tau) d\tau$$

$$\Rightarrow \int_{\tau^2-1}^{\tau^2+1} (t-\tau) d\tau \Rightarrow \underbrace{\int_{\tau^2-1}^{\tau^2+1} t d\tau}_{t \cdot \log(\tau^2+1)} - \underbrace{\int_{\tau^2-1}^{\tau^2+1} \tau d\tau}_{\frac{1}{2} \log(\tau^2+1)}$$

1° caso



$$1) t < 0 \Rightarrow y = 0$$

$$2) t > 0, t-3 < 0 \Rightarrow 0 < t < 3 \Rightarrow \int_0^t \dots$$

$$\begin{aligned} & \begin{matrix} t > \tau \\ 0 < t-\tau < 3 \end{matrix} \Leftrightarrow \begin{matrix} t-3 < \tau \\ t < 3 \end{matrix} \end{aligned}$$

$$3) t-3 > 0 \Rightarrow t > 3 \Rightarrow \int_{t=3}^t \dots \dots$$

$$5) \lambda(t) = \frac{1}{t^2+1} \quad \forall t \quad \text{resoluzioni di: } e^{-d|t|} \Leftrightarrow \frac{2d}{d^2 + 6\pi^2 t^2}$$

$$\Rightarrow \frac{2d}{6\pi^2} \Rightarrow \frac{d}{3\pi^2} = 1 \Rightarrow d = 3\pi^2 \quad e^{-d|t|} \Leftrightarrow \frac{2d}{d^2 + 6\pi^2 t^2}$$

$$\left(\frac{d}{3\pi^2}\right)^2 + t^2 \Rightarrow \frac{1}{j\pi\sqrt{1+t^2}} \Leftrightarrow e^{-e\pi|f|} \pi \quad \text{non sempre possibile}$$

$$\lambda(t) = \frac{1}{(t+i)(t-i)} = \frac{A}{(t+i)} + \frac{B}{(t-i)} \Rightarrow A = -\frac{1}{2i} \quad B = \frac{1}{2i}$$

resoluzioni espresse

$$\Rightarrow \frac{1}{j\pi t} \Rightarrow \frac{d}{\pi} = 1 \Rightarrow \frac{1}{j\pi} \Rightarrow j\pi e^{2\pi f} \ln|-f|$$

$$\frac{d}{j\pi} + t \quad -j + t$$

$$c) \frac{A}{t+\beta} \quad \beta \text{ reale} \Rightarrow \frac{A}{t+t_0} \quad (\text{involuzione temporale})$$

$$-s: \text{pm}(f) = \frac{1}{j\pi t} \Rightarrow -j\pi \text{spm}(f) = \frac{1}{t}$$

$$\text{da } t=t_0 \Rightarrow \lambda(f) e^{j\pi f t_0}$$

$$\Rightarrow \frac{1}{t+d+j\beta} = \frac{1}{\underbrace{(t+d)}_{\text{Re}} + \underbrace{j\beta}_{\text{Imaginaria}}}$$